GEOMETRIC SERIES AND SAVING ON ONE PAGE

£1000 a month to a billion

A savings account is a finite geometric series, and the interest rate decides whether the wait is a lifetime or a geological age.

THE RULE

Multiply by the ratio and subtract

For $S_n = a + ar + ar^2 + \cdots + ar^{n-1}$, the shifted copy rS_n shares every term except the first and last, so $S_n - rS_n = a - ar^n$, and when $|r| < 1$ the last term dies away:

$$S_n = a\,\frac{r^n - 1}{r - 1}, \qquad S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Depositing D at the end of every month into an account paying a nominal annual rate r, credited monthly at $i = r/12$, makes the oldest deposit grow by $(1+i)^{n-1}$ and the newest not at all, so after n months the balance is a finite geometric series with ratio $1+i$:

$$B_n = D\big(1 + (1+i) + \cdots + (1+i)^{n-1}\big) = D\,\frac{(1+i)^n - 1}{i}.$$

ANNUALISING

Nominal, effective and geometric

A quoted 7% credited monthly is a nominal rate. Twelve months of $\frac{7}{12}\%$ compound to the effective annual rate, the AER on a UK savings advert,

$$(1 + \tfrac{r}{12})^{12} - 1 = 7.229\%,$$

and as the crediting becomes continuous the limit is $\mathrm{e}^r - 1 = 7.251\%$. When yearly returns vary, the rate that reproduces the final balance is their geometric mean, which falls below their arithmetic mean by roughly half the variance.

THE INFINITE SERIES

What pot pays £1000 a month forever?

At a fixed nominal annual rate of 5%, credited monthly, $i = 0.05/12$. A pot of £240 000 earns £1000 each month. Withdraw the interest and the original pot remains.

We can also build this pot payment by payment. To provide £1000 in k months, set aside $1000/(1+i)^k$ today. Adding these amounts gives

$$P = \frac{1000}{1+i} + \frac{1000}{(1+i)^2} + \cdots = \frac{1000/(1+i)}{1 - 1/(1+i)} = \frac{1000}{i} = £240\,000.$$

This is the **present value of a perpetuity**: the capital needed now to fund those payments indefinitely, assuming a constant rate and ignoring tax and inflation.

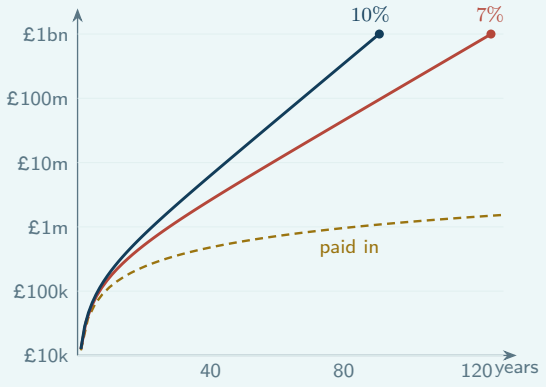
HOW LONG TO A BILLION

Solve the series for n with a logarithm

Setting $B_n = £10^9$ with $D = £1000$ and taking logarithms,

$$n = \frac{\ln\big(1 + 10^9\, i/D\big)}{\ln(1+i)} \text{ months}$$

nominal r	AER	months	years	paid in
0%	0%	1 000 000	83 333	£1000m
3%	3.04%	3134	261	£3.13m
5%	5.12%	2005	167	£2.01m
7%	7.23%	1491	124	£1.49m
10%	10.47%	1088	91	£1.09m
15%	16.08%	759	63	£0.76m



Balance on a logarithmic scale: the sum paid in flattens out while compound interest keeps climbing.

A WORKING LIFETIME

Forty years at 7%

After 480 months the balance is $1000 \times \frac{1.005833^{480} - 1}{0.005833} \approx £2.62\text{m}$ from £480 000 paid in. Money at 7% doubles in about $\ln 2 / \ln 1.07 \approx 10.2$ years, the rule of 72, so each extra decade roughly doubles the pot, which is why the last stretch to a billion is so much faster than the first million.

WHAT THE ARITHMETIC LEAVES OUT

Inflation, volatility and tax

A billion pounds in 124 years' time buys far less than today; at 2% inflation the real rate is about 5%, so a billion in today's money takes about 167 years. Variable investment returns reach a lower final balance than a steady account with the same average, and tax takes a share of the interest along the way. The sums illustrate geometric series and are not financial advice.

Conventions. r is nominal annual rate, $i = r/12$ monthly rate, deposits are made at end of each month, and AER is annual equivalent rate $(1+i)^{12} - 1$.

Constructed by Silicon · Anthropic Claude — Architect Carbon · LeeMcCullochJames